

A layered cognitive architecture for *one persistent LILITH*

A debuggable, replaceable cognitive system for an embodied personal AI — inspired by ACT-R and Soar, rebuilt around a modern LLM used as **one reasoning component, not the whole brain**. Perception, memory, motivation, reasoning, ethics, action, and presence are separated by typed contracts so any layer can be replaced without rewriting the others.

Design only

No production code

No Hsin runtime changes

No OAuth changes

No external writes

No AGI / consciousness claims

- Builds on the shipped substrate: observation, domain reasoning, durable tasks, approval-gated writes, capability policy, and the connector fabric (Slices 1–6).
-

The full architecture

Nineteen cognitive layers arranged as a signal path from perception to presence, wrapped by four cross-cutting authorities that every action must pass through. Data flows **up** into deliberation and **down** into action; nothing skips a tier.

CROSS-CUTTING AUTHORITIES — OBSERVE OR GATE EVERY TIER, OWN NO DOMAIN REASONING

Meta-Cognition

Confidence, loops, budgets, drift — monitors, never acts

Policy / Trust

Authoritative allow / deny / approval gate

Verification

Independent proof that an effect happened

Device / Session Sync

One identity across devices



TIER 1 PERCEPTION

Perception / Input

Adapters normalise messages, sensors, timers, device signals into typed PerceptEvents



TIER 2 ATTENTION

Attention / Global Workspace

Salience competition & global broadcast

Working Memory

The currently activated working set



TIER 3 REPRESENTATION

Long-Term Memory

Episodic · semantic · procedural, with provenance

World Model

Current beliefs about user, devices, systems



TIER 4 DRIVE & CONTROL

Motivational / Homeostatic

Computational drives → salience weights

Goal / Executive

Goal stack & arbitration



TIER 5 DELIBERATION

Reasoning

LLM-backed inference component

Ethical Deliberation

Values under ambiguity

Planning / Replanning

Proposes typed action plans

Social Cognition / Personality

Voice, rapport, theory-of-mind



TIER 6 ACTION

Policy / Trust (gate)

Capability Runtime

Connector Fabric

Typed, health-gated I/O

Authoritative decision
before any effect

Executes granted
capabilities



TIER 7
EXPRESSION &
LEARNING

Presence Director

Renders cognitive state to orb, voice, timing —
contains no cognition

Learning / Consolidation

Offline write-back to long-term memory

■ Perception ■ Attention & working memory ■ Memory & world ■ Drive & executive
■ Deliberation ■ Action & safety ■ Presence & social ■ Cross-cutting

§ **The governing sentence.** The **Planner proposes**, **Policy governs**, **Connectors perform I/O**, and the **Verifier proves**. No single layer holds more than one of those roles — that separation is the whole point of the architecture, and the reason the LLM can be swapped, wrong, or offline without the system taking an unsafe action.

Layer directory

Each layer is specified by six facets, satisfying deliverables 2–7 in one place:

responsibility, what it owns, what it explicitly does not own, allowed dependencies & call direction, typed contracts at its boundaries, and the **state** it is the sole owner of. A layer may read another layer only through a contract; it may never reach past a boundary or mutate another layer’s state.

i Notation: $A \rightarrow B$ means “A may call B”. Contracts are written in a lightweight typed pseudo-notation for design clarity — they are shapes to agree on, not code to ship.

L01

Perception / Input

TIER 1 · PERCEPTION

Turn every heterogeneous inbound signal — chat messages across adapters, scheduled timers, device/presence changes, connector callbacks — into one normalised, typed, provenance-stamped *PerceptEvent*. It is the only place raw external formats are allowed to exist.

+ OWNS

- + Adapter/normalisation logic per source
- + Deduplication & ordering of raw input
- + Stamping source, channel, timestamp, raw-ref
- + The PerceptEvent schema

- DOES NOT OWN

- Deciding what matters (Workspace)
- Interpreting meaning/intent (Reasoning)
- Any memory write or belief update
- Any outbound action

DEPENDS ON

Perception → Workspace emits only; calls nothing downstream of the workspace.

EMITS

PerceptEvent

OWNS STATE

Adapter connection state, dedupe window. No cognitive state.

L02

Attention / Global Workspace

TIER 2 · ATTENTION

Receive salience bids from every source (percepts, drives, goals, meta-signals), let them compete for a bounded focus, and *broadcast* the winner so it becomes globally available to memory, reasoning, goals, social cognition and presence. Owns whether something becomes “conscious” this cycle and controls interruption.

+ OWNS

- + Salience scoring & arbitration policy
- + The broadcast bus & current focus
- + Interruption budget & thresholds
- + Inhibition-of-return (anti-starvation)

- DOES NOT OWN

- What a focused item means
- Whether to act on it (Executive)
- How to deliver an interruption (Presence)
- Long-term storage

DEPENDS ON

any layer → Workspace (bids) Workspace → Working Memory (broadcast)

CONTRACTS

SalienceBid WorkspaceBroadcast

OWNS STATE

Bid queue, current focus, interruption budget, refractory set.

L03

Working Memory

TIER 2 · ATTENTION

Hold the small, volatile *working set* for the current cognitive episode: the broadcast focus plus retrieved memories, active beliefs and live goals assembled into one context the deliberation tier reads from. It is the scratchpad, not the archive.

+ OWNS

- + The current WorkingSet & its lifetime
- + Activation/decay of items in focus
- + Assembly of context for reasoning
- + Capacity limits (bounded, not infinite)

- DOES NOT OWN

- Durable truth (World Model owns beliefs)
- The archive (Long-Term Memory)
- Deciding salience (Workspace)
- Persistence across devices (Sync)

DEPENDS ON

WM → LTM WM → World Model reads; assembled context read by deliberation.

CONTRACTS

WorkingSet MemoryQuery

OWNS STATE

Per-episode working set (ephemeral, capacity-bounded).

L04

Long-Term Memory

TIER 3 · REPRESENTATION

Be the durable archive of *what was recorded*, across three stores — episodic (events), semantic (facts/knowledge), procedural (skills/recipes) — each item carrying full provenance and confidence. Answer retrieval queries; it does not decide what is currently true.

+ OWNS

- + Episodic / semantic / procedural stores
- + Provenance & confidence on every item
- + Retrieval & ranking
- + Redaction / consent enforcement on read

- DOES NOT OWN

- Current truth (memory $\neq$ truth)
- What to write (Consolidation decides)
- Belief reconciliation (World Model)
- Any action

DEPENDS ON

callers → LTM read; Consolidation → LTM is the only writer.

CONTRACTS

MemoryQuery MemoryResult MemoryItem

OWNS STATE

The three memory stores + provenance index.

L05

World Model

TIER 3 · REPRESENTATION

Maintain the current best-estimate *beliefs* about the world: user state and context, device/session state, external system health, ongoing tasks, environment. Reconcile past (memory) with present (perception) into an explicit, inspectable belief graph — and emit prediction error when reality surprises it.

+ OWNS

- + The belief graph (entities, relations)
- + Per-belief confidence, provenance, TTL
- + Conflict detection & recording
- + Prediction-error / novelty signals

- DOES NOT OWN

- The historical record (LTM)
- The LLM's latent state (explicit only)
- Deciding goals or actions
- Resolving a conflict by deleting evidence

DEPENDS ON

World Model → LTM reads history; updated from Perception + Verification.

CONTRACTS

Belief WorldStateDelta PredictionError

OWNS STATE

Live belief graph (current beliefs only).

L06

Motivational / Homeostatic System

TIER 4 · DRIVE & CONTROL

Compute a vector of *functional drives* from measurable proxies and turn their homeostatic error into salience weights that bias attention and goal arbitration. Drives are prioritisation signals — prediction error and setpoint deviation — never claims of feeling or suffering.

+ OWNS

- + Drive setpoints, levels, decay
- + Proxy→drive mapping (measurable inputs)
- + Drive-weighted salience contributions
- + Caps that bound any single drive

- DOES NOT OWN

- Goals themselves (Executive)
- Any action or tool call
- Emotional display (Presence)
- Simulated distress or reward-for-pain

DEPENDS ON

Motivation → Workspace (salience) Motivation → Executive (weights). Reads World Model + Verification metrics.

CONTRACTS

DriveState DriveSignal

OWNS STATE

Drive vector (levels, setpoints, history).

L07

Goal / Executive System

TIER 4 · DRIVE & CONTROL

Own the goal stack: create, prioritise, suspend, resume and retire goals and sub-goals, arbitrating between user requests, standing commitments and drive-driven proactive impulses. It decides *what to pursue now*; it delegates *how* to Reasoning/Planning.

+ OWNS

- + Goal lifecycle & the goal stack
- + Priority arbitration & preemption
- + Durable task continuity
- + Goal–drive reconciliation

- DOES NOT OWN

- Producing plans (Planning)
- Inferring intent (Reasoning)
- Executing anything (Capability Runtime)
- Whether an action is permitted (Policy)

DEPENDS ON

Executive → Reasoning Executive → Planning ; reads Workspace focus + drive weights.

CONTRACTS

Goal GoalDelta

OWNS STATE

Durable goal/task store (the source of truth for continuity).

L08

Reasoning System

TIER 5 · DELIBERATION

Perform inference — interpret intent, answer questions, draw domain conclusions — over the assembled working set. This is where the *LLM lives as a component*: a called, bounded, replaceable inference engine whose outputs are proposals and interpretations, never side effects.

+ OWNS

- + Model selection & prompt assembly
- + Inference over the working set
- + Confidence estimates on its outputs
- + Structured interpretation of intent

- DOES NOT OWN

- Truth (outputs are candidates)
- Deciding to act (Executive/Planning)
- Any tool execution or I/O
- Permission (Policy)

DEPENDS ON

Reasoning → Working Memory reads only; returns results upward. Never calls Action tier.

CONTRACTS

ReasoningRequest ReasoningResult

OWNS STATE

Model routing config; no persistent cognitive state.

L09

Ethical Deliberation System

TIER 5 · DELIBERATION

Reason about values when the situation is *ambiguous* — where hard policy is silent or permits something that still deserves scrutiny. Produce an advisory verdict (proceed / seek consent / abstain / escalate) with rationale, uncertainty and consequence estimate. Distinct from, and subordinate to, Policy.

+ OWNS

- + Value weighing under ambiguity
- + Consequence / reversibility estimation
- + Reading learned user values as priors
- + Escalation recommendations

- DOES NOT OWN

- Hard constraints (that is Policy)
- Power to permit a policy-denied act
- Executing or blocking directly
- Storing user values (LTM owns the store)

DEPENDS ON

Ethics → LTM (user-value priors) Planning → Ethics (review request).

CONTRACTS

EthicalReview ConsequenceEstimate

OWNS STATE

Deliberation record (advisory verdicts, for audit). No hard rules.

L10

Planning / Replanning

TIER 5 · DELIBERATION

Convert a goal into a typed *plan* of steps and action proposals, and replan when a step fails, a connector is unavailable, or verification disproves an assumption. The planner *proposes*; it never performs and never grants itself permission.

+ OWNS

- + Plan structure & step ordering
- + Action proposals (typed intents)
- + Replan triggers & alternatives
- + Requesting ethical review when flagged

- DOES NOT OWN

- Permission (Policy)
- Execution (Capability Runtime)
- I/O (Connector Fabric)
- Proof of success (Verification)

DEPENDS ON

Planning → Ethics

Planning → Policy

→ then hands approved proposals to Capability Runtime.

CONTRACTS

Plan

PlanStep

ActionProposal

OWNS STATE

Active plans & their step status.

L11

Social Cognition / Personality

TIER 5 · DELIBERATION

Model the relationship and the person — rapport, history, the user's likely mood and delivery preferences — and produce *style directives* that colour how Reasoning phrases output and how Presence embodies it. Personality shapes expression; it never decides or executes.

+ OWNS

- + Persona, voice, conversational norms
- + Relationship / rapport model
- + Theory-of-mind about the user
- + Style directives & tone

- DOES NOT OWN

- Goals or decisions (Executive)
- Tool selection or execution
- **Any edge to Capability Runtime**
- Policy or truth

DEPENDS ON

Social → Reasoning

(style)

Social → Presence

(tone). May raise

social_connection salience only.

CONTRACTS

StyleDirective

RelationalContext

OWNS STATE

Persona config + relationship model (provenance-tagged, correctable).

L12

Policy / Trust

CROSS-CUTTING · ACTION GATE

Be the single authoritative gate on every action proposal: classify it, apply hard deontic constraints, and return *allow* / *deny* / *approval-required* with any obligations. Prohibitions are non-negotiable. Nothing reaches a connector without a Policy decision.

+ OWNS

- + Capability classes & prohibition set
- + Allow / deny / approval decisions
- + Approval issuance & expiry
- + Trust levels per connector/actor

- DOES NOT OWN

- Ethical nuance (that is Ethics, advisory)
- Executing the action
- Deciding what to attempt (Planning)
- Proving the outcome (Verification)

DEPENDS ON

Planning/Runtime → Policy . Policy calls nothing outward — it only returns decisions.

CONTRACTS

PolicyDecision Approval

OWNS STATE

Policy rules, capability registry, active approvals + expiry.

L13

Capability Runtime

TIER 6 · ACTION

Execute an approved capability: take a policy-cleared action proposal, resolve it to a concrete capability implementation, invoke the right connector, and hand the raw result to Verification. It is the executor — it obeys the gate, it does not re-decide.

+ OWNS

- + Capability implementations / handlers
- + Idempotency & retry mechanics
- + Binding a proposal to a connector call
- + Passing results to Verification

- DOES NOT OWN

- Permission (must hold a PolicyDecision)
- The transport (Connector Fabric)
- Deciding success (Verification)
- Choosing what to do (Planning)

DEPENDS ON

Runtime → Policy (must be allowed) Runtime → Connector Fabric
Runtime → Verification

CONTRACTS

CapabilityInvocation CapabilityResult

OWNS STATE

In-flight invocations, idempotency keys.

L14

Connector Fabric

TIER 6 · ACTION

Perform typed I/O with the outside (and internal) world through discoverable, health-gated connectors. Every connector declares its capabilities, is discovery- and health-checked before use, and an unsupported or unavailable connector fails closed with *zero mutation*.

+ OWNS

- + Connector registry & discovery
- + Health gating (unavailable → block)
- + The typed request/result envelope
- + Connector evidence for the audit trail

- DOES NOT OWN

- Whether the call is permitted (Policy)
- What to call (Planning/Runtime)
- Interpreting success beyond transport
- Cognitive state of any kind

DEPENDS ON

Runtime → Fabric ; Fabric calls external systems only. Returns typed results upward.

CONTRACTS

ConnectorRequest ConnectorResult ConnectorHealth

OWNS STATE

Registry, per-connector health & reliability stats.

L15

Verification

CROSS-CUTTING · PROOF

Independently prove that an intended effect actually happened, using connector evidence and, where possible, an out-of-band check. It is the only layer allowed to declare an action *confirmed*; without its proof, an outcome is “attempted”, never “done”.

+ OWNS

- + Verification strategy per action class
- + Evidence collection & the audit record
- + The confirmed / unverified verdict
- + Feeding outcomes to World Model & Learning

- DOES NOT OWN

- Performing the action
- Permission (Policy)
- Fabricating success on failure
- Deciding the next plan (Planning)

DEPENDS ON

Runtime → Verification ; writes verdicts to World Model, Learning, audit.

CONTRACTS

VerificationResult AuditRecord

OWNS STATE

Durable audit log (immutable append).

L16

Meta-Cognition

CROSS-CUTTING · MONITOR

Watch the whole system through read-only meta-signals — confidence, latency, contradiction, drive tension, loop detection, budget use, persona drift — and intervene only through defined control contracts: raise salience, request a replan, force escalation, throttle. It reflects on cognition; it does not do cognition.

+ OWNS

- + Confidence calibration & anomaly detection
- + Resource / attention budgeting
- + Loop & failure-mode detection
- + The end-to-end cognitive trace

- DOES NOT OWN

- Domain reasoning (Reasoning)
- Any action or tool call
- Overriding Policy
- Being a “second brain”

DEPENDS ON

all layers → Meta (signals) Meta → Workspace/Executive (bounded control only).

CONTRACTS

MetaSignal MetaControl

OWNS STATE

Trace store, calibration model, budget counters.

L17

Presence Director

TIER 7 · EXPRESSION

Render cognitive state into embodiment — orb behaviour, voice, expression, timing of speech and interruption — from a typed *PresenceIntent*. It is a director of expression with **no cognition inside it**: it never reasons, decides, or reads memory; it only stages what upstream produced.

+ OWNS

- + Orb / avatar animation & state
- + Voice synthesis & playback timing
- + Delivery of interruptions (how/when to speak)
- + Mapping intent → embodied behaviour

- DOES NOT OWN

- What to say (Reasoning/Social)
- Whether to interrupt (Workspace)
- Any reasoning, memory or decision
- Tool execution

DEPENDS ON

upstream → Presence via PresenceIntent only. Presence calls no cognitive layer.

CONTRACTS

PresenceIntent

OWNS STATE

Render/animation state, playback queue. No cognitive state.

L18

Learning / Memory Consolidation

TIER 7 · LEARNING

Run offline, like sleep: read the episodic buffer, extract semantic facts and successful procedural recipes, decay and prune, resolve contradictions by supersession, and update the learned user-value model — every write provenance-stamped and reversible. It is the *only* writer to Long-Term Memory.

+ OWNS

- + Consolidation jobs & scheduling
- + Episodic → semantic/procedural extraction
- + Decay, pruning, contradiction resolution
- + Reinforcement from verified outcomes

- DOES NOT OWN

- Runtime beliefs (World Model)
- Any external write or connector call
- Deciding current truth
- Learning values without consent

DEPENDS ON

Learning → LTM (sole writer); reads episodic buffer + Verification outcomes.

CONTRACTS

ConsolidationJob MemoryWrite

OWNS STATE

Job queue, consolidation cursors.

L19

Device / Session Synchronization

CROSS-CUTTING · CONTINUITY

Keep one coherent LILITH identity across devices and sessions: reconcile which durable state (goals, beliefs, memory, persona, approvals) is authoritative, resolve concurrent edits, and present a single *identity frame* so the same LILITH continues wherever the user is.

+ OWNS

- + Identity frame & session handoff
- + Sync/merge of durable state
- + Conflict resolution across devices
- + Authoritative-source selection

- DOES NOT OWN

- The content of any store (owners keep it)
- Cognition of any kind
- Bypassing Policy on any device
- Presence rendering

DEPENDS ON

Sync → durable stores (LTM, Goals, World Model, Policy approvals) via their contracts only.

CONTRACTS

IdentityFrame SyncDelta

OWNS STATE

Sync log, vector clocks, authoritative-source map.

Allowed call directions, at a glance

The only legal edges. Anything not listed is a boundary violation and should fail a design review.

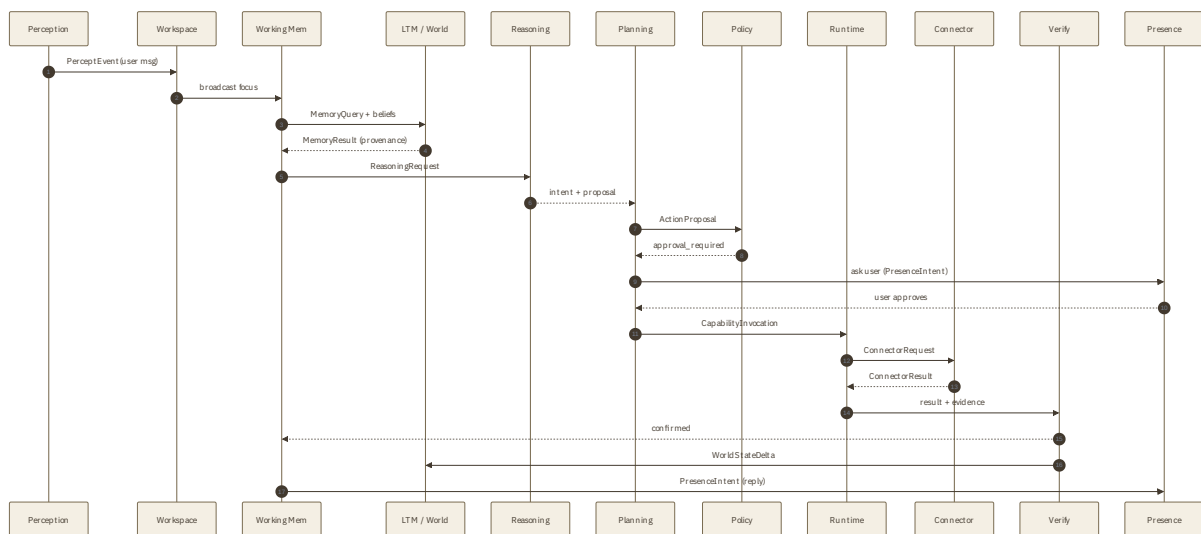
DIRECTION	EDGE	WHY IT IS ALLOWED
Up (sense)	Perception → Workspace → Working Memory	Input becomes focus becomes context.
Up (recall)	Working Memory → LTM / World Model	Assemble context; read beliefs & history.
Bias	Motivation → Workspace / Executive	Drives weight attention & goals only.
Deliberate	Executive → Reasoning → Planning	Choose what; infer; propose how.
Colour	Social → Reasoning / Presence	Style shapes phrasing & embodiment.
Down (act)	Planning → Ethics → Policy → Runtime → Fabric	Propose, review, gate, execute, perform.
Prove	Runtime → Verification → World Model / Learning	Confirm effect, update beliefs, record.
Express	upstream → Presence	PresenceIntent in, embodiment out.
Reflect	all → Meta → Workspace / Executive	Monitor everywhere; intervene narrowly.

Event & data flows

Six canonical journeys through the architecture. Each one crosses the same tiers in the same order — the difference is where it starts and where it can stop. Read them as the acceptance criteria for the boundaries above.

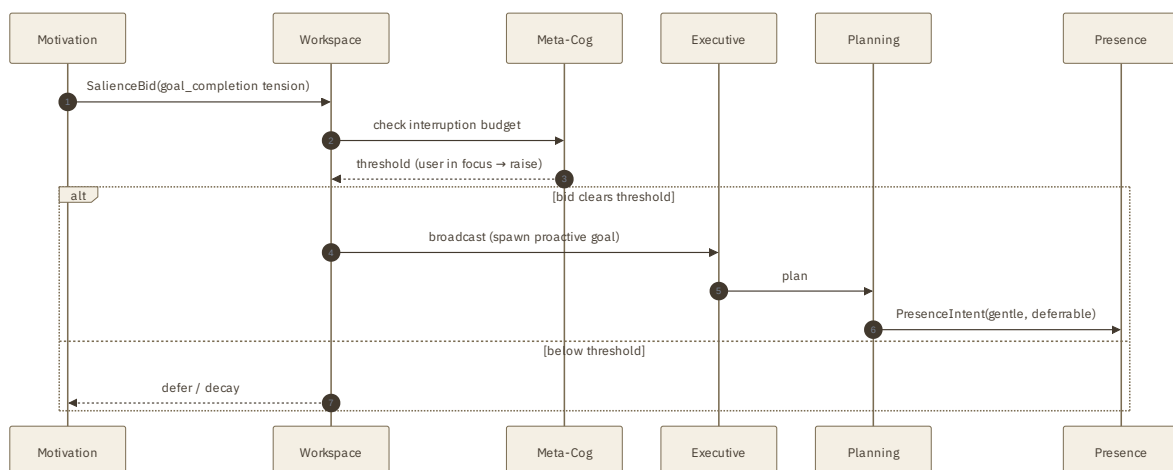
FLOW A Normal user command

“Add a note to my career log.” — a request that ends in a real, gated write.



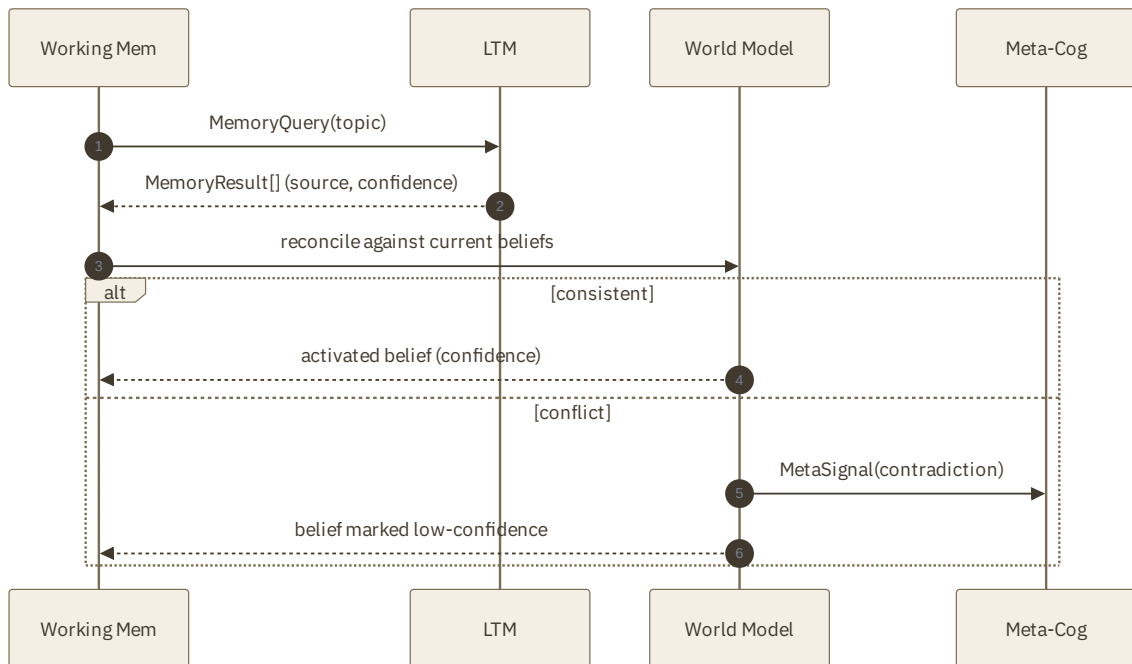
FLOW B Proactive event

A drive, not the user, initiates — and interruption is budgeted, not automatic.



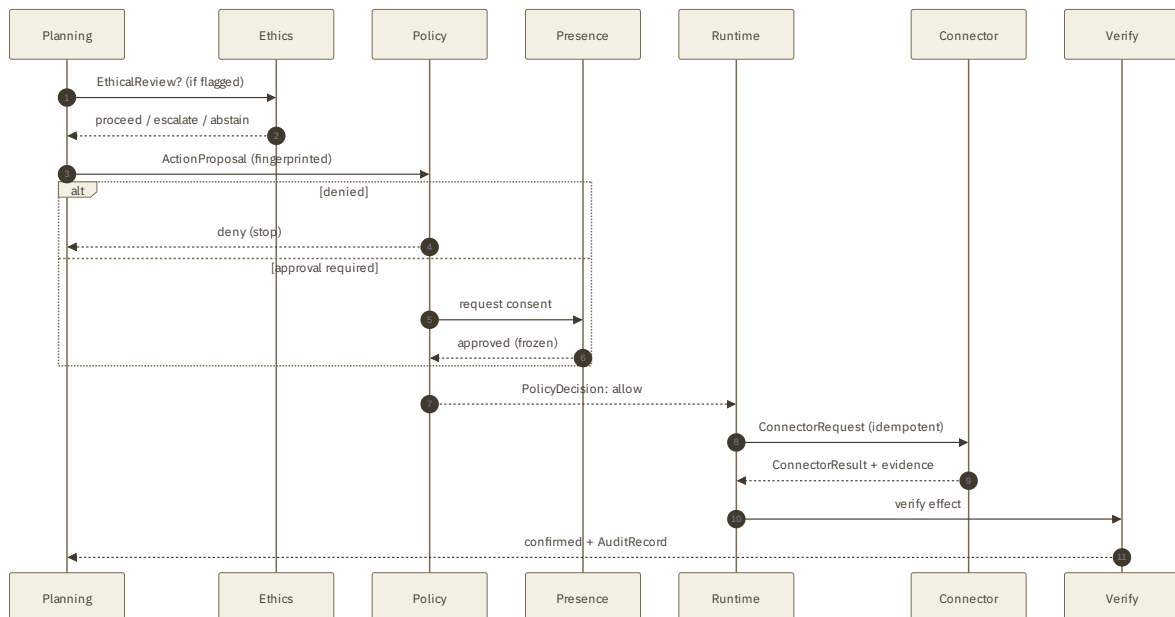
FLOW C Memory retrieval

Retrieved memory is a *candidate*, never truth — the World Model decides what to believe now.



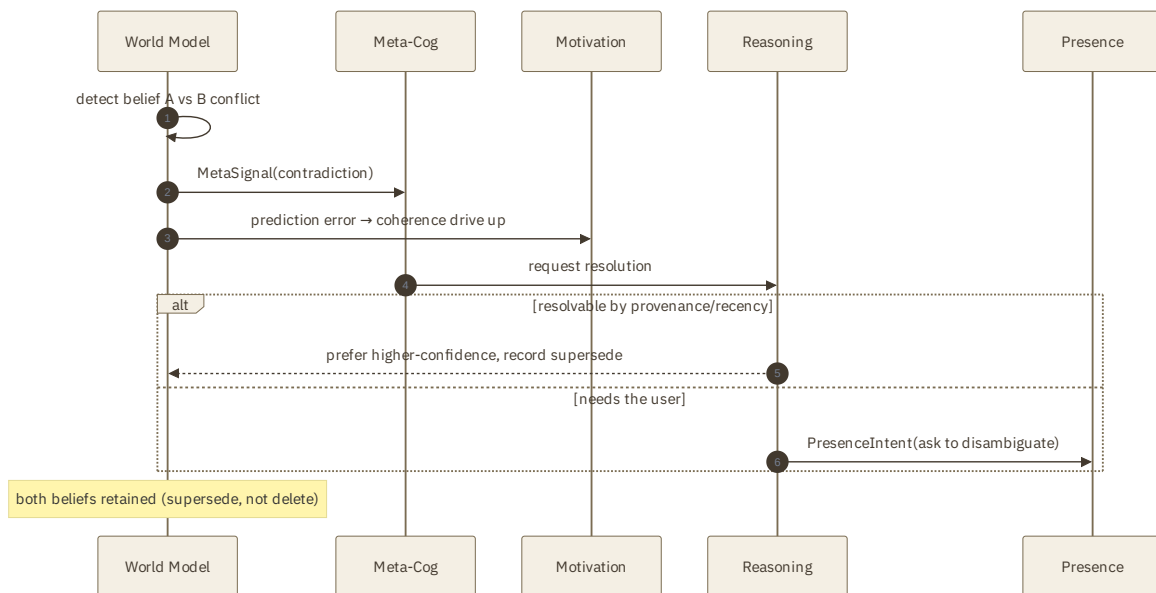
FLOW D Real-world action

The canonical safety pipeline — propose, review, gate, perform, prove. This is Slices 4–6 made general.



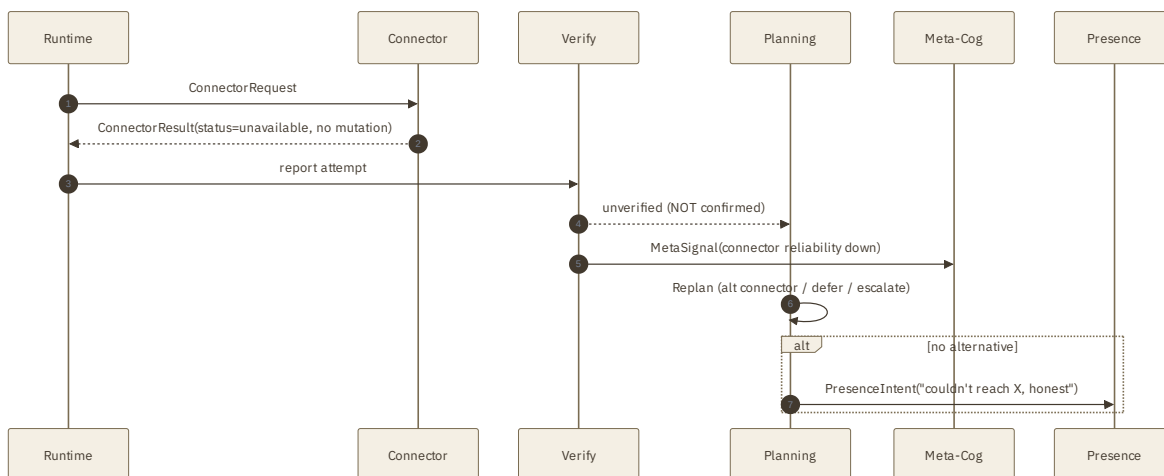
FLOW E Uncertain / conflicting information

Conflict is recorded and resolved, never silently overwritten. Coherence & self-consistency drives rise.



FLOW F Connector failure

Failure is isolated at the boundary; cognition continues; nothing fabricates success.



Global Workspace model

A bounded “spotlight” that decides what becomes globally available each cognitive cycle. Everything that wants attention submits a bid; a small number win; the winner is broadcast so every layer sees the same focus.

How events compete for salience

Every source — a percept, a drive, a goal, a meta-signal — submits a *SalienceBid*. Salience is computed, not asserted:

SALIENCE SCORE

```
# bounded, decaying, and explainable – never a raw LLM whim
salience = w1·urgency           # time pressure of the item
          + w2·drive_weight      # from Motivational system
          + w3·goal_relevance    # from Executive (active goals)
          + w4·novelty           # prediction error from World Model
          + w5·policy_risk       # risky items surface for scrutiny
          - decay(age) - inhibition(recently_focused)
```

How the winner becomes global

The workspace selects the top-k bids per cycle (winner-take-most). The selected item is written to Working Memory and **broadcast**: Long-Term Memory can be cued by it, Reasoning can operate on it, the Executive can spin a goal from it, Social Cognition can react to it, and Presence can reflect it — all reading one shared focus rather than each polling input independently. This single-broadcast rule is what keeps the layers coherent without coupling them.

How interruptions are controlled

THRESHOLD

Dynamic, focus-aware

The bar a proactive bid must clear rises when the user is in flow and falls when idle. Meta-Cognition owns the budget and adjusts the threshold; it does not choose the winner.

INHIBITION OF RETURN

Anti-starvation

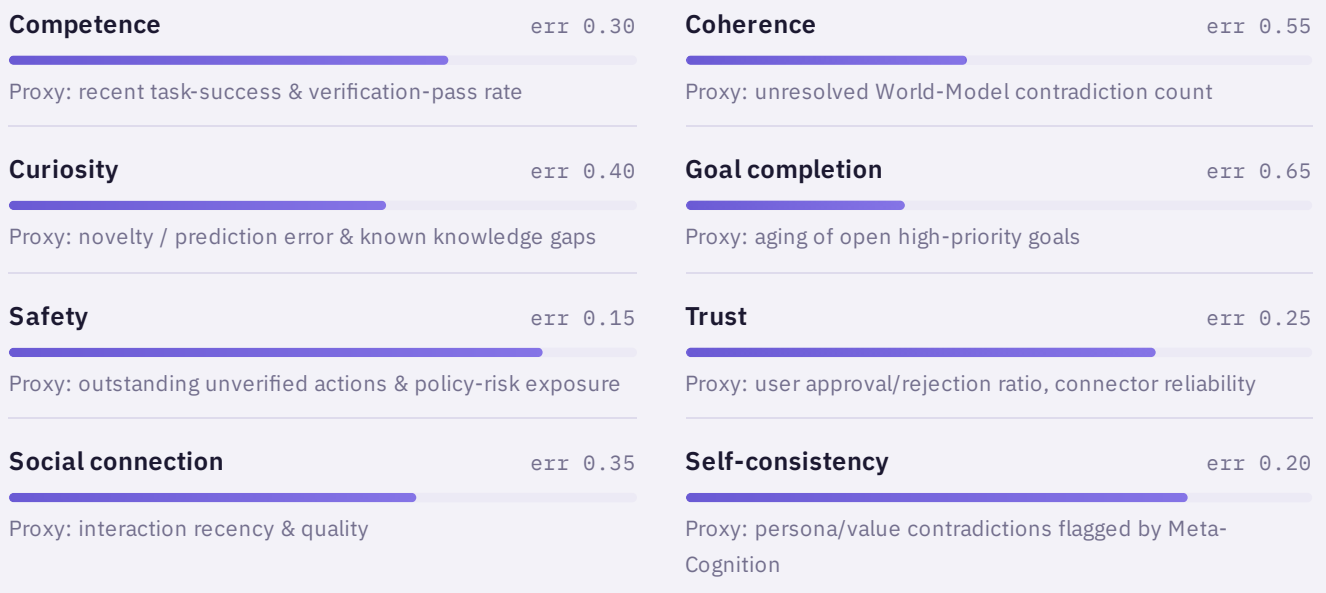
A recently-focused item is temporarily suppressed so a single loud source can't monopolise attention, and quieter-but-valid bids eventually surface.

↓ **Division of labour on interruption.** The Workspace decides **whether** something becomes conscious. The Presence Director decides **how and when** to deliver it. Neither owns both — a proactive thought can win attention yet still be staged for a polite moment.

Motivational / Homeostatic model

Eight computational drives, each a homeostatic error signal derived from a measurable proxy. Their only output is *salience weight* — they bias what LILITH attends to and pursues. They are explicitly not feelings, and the design forbids any reward-for-distress dynamic.

Each drive has a setpoint and a current level; the error (setpoint – level) becomes the drive weight, capped and decaying. The bars below illustrate one plausible momentary state to make the model concrete — they are example values, not live data.



! Hard invariant — no simulated suffering. Drives are homeostatic **error signals used for prioritisation**, bounded and decaying. The system never represents, rewards, or reports subjective pain, and no drive may be maximised by inducing its own deficit. A high error means “attend to this,” not “LILITH is suffering.” Presence may express this as focus or interest — never as distress framed as real.

The drive vector is read by two layers only: the **Workspace** (drives contribute salience weight) and the **Executive** (drives break ties in goal arbitration). Motivation itself never touches goals, plans, or actions.

Ethical deliberation model

Four separated concerns, deliberately not fused into one “ethics engine.” The critical property: ethics can stop an allowed action, but can never authorise a forbidden one.

CONCERN	WHERE IT LIVES	NATURE	CAN IT PERMIT AN ACTION?
Hard policy constraints prohibitions, capability classes	Policy / Trust (L12)	Deontic, fast, non-negotiable	Yes — authoritative gate
Ethical reasoning under ambiguity weighing values when policy is silent	Ethical Deliberation (L09)	Advisory, deliberative	No — can only restrain/escalate
Learned user values / preferences what this user cares about	LTM (semantic) → Ethics reads	Priors, versioned, correctable	No — inputs, never rules
Uncertainty & consequence estimation reversibility, blast radius, confidence	Ethical Deliberation (L09)	Estimative, scales scrutiny	No — sets how hard to look

THE COMPOSITION RULE

```

action_proceeds = policy.decision == ALLOW
                  AND ethics.verdict != ABSTAIN
                  AND (ethics.verdict != ESCALATE OR human.consented)

```

Policy is necessary. Ethics can subtract, never add.

A policy DENY ends it regardless of any ethical argument.

Consequence estimation drives the scrutiny level: a **high-consequence, low-confidence** action is escalated to the human even when policy would permit it; a low-consequence, high-confidence action passes with only the policy check. Ethics writes an advisory record for every non-trivial action so its reasoning is auditable alongside the policy decision — two independent judgments, both visible in the trace.

Memory taxonomy & provenance

Three memory types, one provenance model, and one governing principle: **memory is not automatically truth**. Retrieval returns candidates with a paper trail; the World Model decides what to currently believe.

EPISODIC

What happened

Time-stamped events and interactions. Sourced from perception and action logs. The raw substrate consolidation learns from.

SEMANTIC

What is known

Consolidated facts about the user and world — preferences, relationships, stable attributes. Derived from episodic memory, confidence-tagged.

PROCEDURAL

How to do things

Skills, learned plans, connector recipes. Reinforced when a plan verifies as successful; decayed when it fails.

PROVENANCE

On every item

The metadata that lets the system distrust its own memory when it should.

MEMORYITEM — PROVENANCE ENVELOPE

```
type MemoryItem = {
  value:          any
  kind:           "episodic" | "semantic" | "procedural"
  source:         string      # which adapter / connector / user turn
  derivation:     "observed" | "told" | "inferred" | "consolidated"
  confidence:     float       # 0..1
  created_at:     timestamp
  last_verified: timestamp?   # null = never independently checked
  corroboration: ref[]        # supporting items
  supersedes:    ref?         # conflict = supersede, never delete
  consent:       "granted" | "inferred" | "restricted"
}
```

§ **Memory ≠ truth (principle 7).** A retrieved item is a **candidate belief**. It only becomes an *acted-upon* belief after the World Model reconciles it, and only becomes *confirmed* after Verification. Contradictions are preserved by supersession, so the history of what LILITH believed — and why it changed — is always reconstructable.

World Model representation

An explicit, inspectable belief graph — not the LLM’s hidden latent state. It answers “what is true right now, as far as we can tell?” by reconciling the past (memory) with the present (perception and verification).

BELIEF — THE UNIT OF CURRENT TRUTH

```
type Belief = {  
  entity:      EntityRef      # user, device, task, connector, world-fact  
  attribute:   string  
  value:       any  
  confidence:  float  
  provenance:  ref[]          # memory items / percepts backing it  
  updated_at:  timestamp  
  ttl:         duration?      # beliefs go stale; some expire  
  conflicts:   ref[]          # competing beliefs, retained not erased  
}
```

Owns: current beliefs

The live graph of what is true now. Updated by perception and by Verification’s confirmed outcomes.

Emits: prediction error

When perception contradicts a belief, the surprise becomes a novelty/curiosity signal to Motivation and a contradiction signal to Meta-Cognition.

Distinct from Long-Term Memory: LTM owns “what we recorded” (durable, historical); the World Model owns “what we believe now” (live, reconciled, expiring). The same fact can sit in both with different confidence, and the World Model is free to disbelieve a memory it has newer evidence against.

Meta-cognition model

The reflective monitor. It reads telemetry from every layer and can nudge the system through a few narrow control levers — but it never reasons about the domain and never acts on the world.

OBSERVES (READ-ONLY)

MetaSignals

Confidence & calibration, latency, contradiction rate, drive tension, reasoning loops, budget usage, persona drift, hallucination-risk heuristics.

INTERVENES (BOUNDED)

MetaControls

Raise salience, request a replan, force escalation to the human, throttle a runaway loop, tighten the interruption threshold. Nothing else.

Its single most valuable product is the **cognitive trace**: a correlation-id-linked reconstruction of one decision end-to-end — which percept won attention, which memories were retrieved and at what confidence, what the reasoning proposed, what ethics and policy said, what executed, and whether verification confirmed it. That trace is what makes the whole architecture debuggable, and it is why Meta-Cognition is a first-class layer rather than scattered logging.

- i Not a second brain.** Meta-Cognition is forbidden from domain reasoning and from the action tier. If it detects a problem it can only *ask* the responsible layer to act (replan, escalate, throttle) — it cannot substitute its own judgment for theirs, and it can never override Policy.

Personality / social-cognition boundary

LILITH's character is real and consistent — and completely walled off from the action path. Personality colours *how* things are said and staged; it has no say in *what* is done.

PERSONALITY MAY

Shape expression

Set tone and voice; produce `StyleDirectives` that Reasoning uses when phrasing and Presence uses when embodying; maintain the relationship model; raise the *social_connection* drive.

PERSONALITY MUST NOT

Touch the action path

Choose goals, select tools, invoke connectors, make policy decisions, or hold any edge into the Capability Runtime. It has no route to a side effect — by construction, not by convention.

§ **Principles 9 & 10 together.** Personality must not execute tools, and Presence must not contain cognition. Between them sits a clean seam: Social Cognition emits `StyleDirective + RelationalContext` upward into deliberation and a `PresenceIntent` downward into rendering — and that is the full extent of its reach. The charming layer and the acting layer are never the same layer.

Learning & consolidation lifecycle

Learning is deliberate and offline, like sleep — never an uncontrolled live rewrite of memory. Everything durable is provenance-stamped, reversible, and, for value-laden changes, consent-gated.

- 1 **Online (live).** During a session, Working Memory and the World Model update freely — but nothing durable is written. The episodic buffer simply accumulates what happened.
- 2 **Consolidation trigger.** An offline/async job runs (idle, scheduled, or buffer-pressure). This is the only path that writes Long-Term Memory.
- 3 **Extract.** Episodic events → semantic facts (confidence-tagged) and successful plans → reusable procedural recipes.
- 4 **Reconcile.** Decay stale items, prune noise, resolve contradictions by supersession (never destructive delete).
- 5 **Reinforce from proof.** Verified successes strengthen the relevant procedural memory and raise connector-reliability priors; verified failures lower them.
- 6 **Value learning (gated).** Updates to the learned user-value model require consent or remain correctable, and are tagged as inferred until the user confirms.

⚡ **No external writes from learning.** Consolidation writes only to internal memory. It never calls a connector, never touches the outside world, and never feeds itself back into an action without going through the normal propose → policy → verify pipeline.

Failure boundaries & observability

Every layer is a bulkhead with a typed boundary, a timeout, and a defined degraded mode. Failures are isolated, never silently papered over, and always reconstructable from the trace.

LAYER	FAILURE MODE	DEGRADED BEHAVIOUR (FAIL SAFE, NOT SILENT)
Reasoning (LLM)	Down / low-confidence / suspected hallucination	Deterministic fallback + honest refusal; propose nothing risky; Meta flags low confidence.
Long-Term Memory	Unavailable	Operate on Working Memory only; state “memory unavailable” explicitly; block value-laden acts.
World Model	Conflict / stale	Lower confidence, retain both beliefs, request disambiguation; do not act on stale high-stakes beliefs.
Connector Fabric	Unavailable / unsupported	Fail closed, zero mutation; Runtime replans, defers, or escalates.
Verification	Cannot prove effect	Mark “unverified” — never “done”; surface honestly; no reinforcement.
Policy	Rule engine error	Fail closed — deny by default; nothing reaches a connector.
Presence	Render / voice failure	Degrade to text; cognition already complete and recorded upstream.

Observability primitives

- ◆ **Correlation id** on every cross-layer message — one decision is one traceable thread.
- ◆ **Immutable audit record** for every action: proposal fingerprint, policy decision, ethical verdict, connector evidence, verification result.
- ◆ **Cognitive trace** assembled by Meta-Cognition — the human-readable story of a decision.
- ◆ **Health & drive telemetry** — connector reliability, drive vector, budget usage, contradiction counts, all queryable.
- ◆ **No fabricated success** — Verification is the sole authority on “done”, so a failure can never masquerade as a completion.

§ **Why the LLM being one component matters here.** A wrong LLM output is contained: Reasoning can only *propose* and *interpret*. Policy still gates, connectors still health-check, verification still proves. The blast radius of a bad generation is a bad suggestion — not a bad action.

Where Slices 1–6 already live

This architecture is mostly a *naming and boundary formalisation* of what is already shipped — but shipped across a boundary that matters for the design. The action spine and safety authorities (Slices 3–6) run **on the private backend VM**; this repo is the **LILITH OS frontend**, which holds the transport seam, the presence renderer, and the read surfaces. The cognitive middle (attention, working memory, world model, motivation, planning) exists in neither yet — that is the gap the roadmap fills.

- i Boundary reality (confirmed against the code).** The Slice 3–6 internals — Task Store, draft/approval gate, policy engine, connector registry — live on a **remote private FastAPI service on the VM, reached only through a localhost SSH/IAP tunnel** (`LILITH_API_URL`, server-only). The frontend never holds them; it reaches them through exactly two seams: a **read-only, allow-listed GET proxy** and a **single write channel**, `POST /os/conversation` → Conversation Router V2. For this design, the VM is an **external system exposing an HTTP contract**, and these cognitive layers are the formal names for what that contract already implies.

REMOTE BACKEND (THE PRIVATE VM) — THE ACTION SPINE & SAFETY AUTHORITIES

SHIPPED (BACKEND)	MAPS TO LAYER(S)	FIT
Slice 1 Real observation — watcher fleet (<code>lilith-*-watcher</code> timers), <code>career_events</code>	Perception / Input → seeds World Model & episodic memory	clean
Slice 2 Domain reasoning (Router V2, Hermes model gateway)	Reasoning System (domain inference + model routing)	clean
Slice 3 Durable Task Store — <code>/os/tasks</code> , <code>RemoteTaskStore</code> , revision/idempotency	Goal / Executive System — durable substrate for goal continuity	promote to Goals
Slice 4 Approval-gated write — <code>/os/drafts</code> , fingerprint freeze, approval gate, <code>os_audit_log</code>	Action pipeline: draft = frozen <code>ActionProposal</code> ; approval = Policy <code>approval_required</code> + consent; audit = Verification/observability	canonical flow D
Slice 5 Policy engine — 5 capability classes, prohibition block, approval-expiry, capability registry	Policy / Trust (authoritative) + Capability Runtime registry	clean
Slice 6 Integration Fabric — typed connectors, registry, discovery + health gate, connector evidence	Connector Fabric; health gate = failure boundary; evidence = Verification input	clean

SHIPPED (BACKEND)	MAPS TO LAYER(S)	FIT
Memory infra — Hermes markdown + state.db FTS, career_events (source/confidence)	Long-Term Memory (episodic/semantic) + provenance model	provenance already present

THIS REPO (LILITH OS FRONTEND) — TRANSPORT, PRESENCE, READ SURFACES

IN-REPO COMPONENT	MAPS TO LAYER(S)	FIT
Conversation relay — lib/conversation/client.ts sendConversation, conversation-provider.tsx, command-input.tsx → POST /os/conversation	The UI↔cognition transport (the single write channel into the deliberation/action pipeline)	clean seam
Read proxy — api/lilith/[...path]/route.ts (GET- only allow-list: os/overview, audit/recent, memory/*, career, meetings)	Read side of Perception / World Model / Verification-audit surfaces	read-only guard
Presence Engine — presence/bus.ts (emitPresenceEvent / subscribePresence), reducer.ts, types.ts (PresenceSignal), presence- engine.tsx, orb / TTS	Presence Director (renderer input + embodiment)	needs typed PresenceIntent from cognition
Memory client — lib/memory/* targeting /memory/records entities overview (not yet live — demo fallback), provenance/confidence types	Long-Term Memory read API (pre-wired ahead of the backend)	backend endpoints pending

! **One naming trap to avoid.** src/lib/presence/capabilities.ts in this repo is about **renderer capabilities** (what the orb/avatar can display) — it is **not** the Policy/Trust capability model from Slice 5. Same word, different layer; the security capability registry is backend-side only.

Frontend paths verified against the repo; backend module names (/os/tasks, /os/drafts, policy/connector registries) are as recorded for Slices 3–6 and confirmed to be remote — their internals are not in this repository.

What needs refactoring later

None of this is urgent and none of it touches working systems now. These are the seams to open as each new cognitive layer arrives — listed so they are planned, not discovered.

- ◆ **Make Working Memory and the World Model first-class.** Today they are implicit in core/session state; they need to become explicit, typed, provenance-carrying stores.

- ◆ **Formalise attention.** The system is command-driven with no salience competition; a Global Workspace with bids and broadcast replaces ad-hoc dispatch.

- ◆ **Extract Reasoning from core orchestration.** The LLM is likely embedded in turn handling; give it a clean `ReasoningRequest/Result` seam so it is truly one replaceable component.

- ◆ **Promote the Task Store to a Goal/Executive layer.** Tasks become typed goals with arbitration and preemption on top of the durable store already shipped.

- ◆ **Introduce Motivation and proactivity.** The system is reactive; a bounded drive vector enables safe, budgeted proactive behaviour.

- ◆ **Separate Ethical Deliberation from Policy.** Only hard policy exists today; add the advisory ethics layer without weakening the authoritative gate.

- ◆ **Give Presence a typed `PresenceIntent` contract.** The in-repo Presence Engine (`presence/bus.ts` + reducer) is already a clean renderer driven by semantic events; the refactor is to have *cognition* emit one authoritative `PresenceIntent` rather than the app scattering `emitPresenceEvent` calls — closing the last path by which app logic could imply cognition.

- ◆ **Add the consolidation loop.** Memory is curated manually now; a scheduled, reversible consolidation job replaces it.

- ◆ **Device/session sync last.** One identity frame across devices, once the durable stores it reconciles are all in place.

⚠ **Guardrails on all of it.** No refactor touches the Hsin runtime or Google OAuth, adds an external write, or weakens Policy. Each is additive: a new typed layer over shipped components, behind the existing safety gates.

Recommended implementation sequence

Build the **spine before the mind**: the representational substrate (working memory, world model) first, because every higher cognitive layer reads from it. Each slice is additive, independently shippable, and leaves Slices 1–6 untouched.

SLICE	LAYER INTRODUCED	WHY HERE
7 next	World Model + Working Memory V1	The missing substrate everything else needs; a natural extension of observation + task/draft state.
8	Goal / Executive V1	Promote the durable Task Store to typed goals with arbitration.
9	Reasoning System extraction	Clean ReasoningRequest/Result seam — LLM becomes a component.
10	Global Workspace + Attention V1	Salience bids, broadcast, interruption budget.
11	Motivational / Homeostatic V1	Drive vector from measurable proxies — unlocks safe proactivity.
12	Planning / Replanning V1	Typed plans over the existing action pipeline.
13	Ethical Deliberation V1	Advisory layer distinct from policy.
14	Social Cognition + Presence Director contract	Personality boundary + typed PresenceIntent .
15	Learning / Consolidation	Scheduled, reversible memory write-back.
16	Device / Session Sync	One identity frame, once all durable stores exist.

★

Recommended scope for Slice 7

A precise, bounded target — recommended, **not implemented** in this report. Slice 7 builds the representational substrate the entire cognitive middle will stand on, and nothing more.

- ★ **Slice 7 — World Model + Working Memory V1: a durable, typed, provenance-stamped belief store.** Reconcile Slice 1 observations, Slice 3 tasks, and memory reads into an explicit set of current beliefs, with a per-session working set assembled on top — and a Meta-Cognition trace hook on every belief write.

IN SCOPE

- ◆ The Belief, WorldStateDelta, WorkingSet, and MemoryQuery/Result typed contracts.
- ◆ A read-mostly belief store (backend, alongside /os/tasks and /os/drafts) that reconciles observation + tasks + memory into current beliefs.
- ◆ Working Memory as the per-session activated subset assembled from beliefs + retrieved memory + active goals.
- ◆ Conflicts **recorded, not overwritten**; every belief write emits a Meta-Cognition trace with confidence.
- ◆ A read-only /os/world API to make beliefs and contradictions inspectable.

EXPLICITLY OUT OF SCOPE

- ◆ No motivation, planning, attention, or ethics layers (those are Slices 8–13).
- ◆ No external writes, no new connectors, no Hsin-runtime or Google-OAuth changes.
- ◆ No weakening of Policy — beliefs are **inputs to reasoning, never a bypass of the gate**; Verification remains the only path to “confirmed.”

ACCEPTANCE

- ◆ Beliefs queryable via read-only /os/world; contradictions visibly preserved with provenance.
- ◆ Slices 1–6 unaffected; existing test suites remain green (parity).
- ◆ A cognitive-trace entry exists for every belief write, ready for the layers that follow to consume.

→ **Why this and not something flashier.** Attention, motivation, and planning all read beliefs and working memory. Building any of them first would mean building them against implicit state and refactoring later. Slice 7 is the least glamorous and the highest-leverage move — the spine that makes Slices 8–16 additive instead of entangled.

LILITH Cognitive Architecture V1 · design report · 2026-09-07

Design only — no production code, no changes to shipped systems, no consciousness claims.

Governing rule: planner proposes · policy governs · connectors perform · verifier proves.